

Nonlinear Solid Mechanics

A COMPILATION OF
NOTES, NOTATIONS, AND
ILLUSTRATIONS

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Abstract

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Acknowledgements

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Chapter 1

Introduction

1.1 Purpose and Scope

- The purpose of this manuscript is to memorialize the theories of solid dynamics, thermodynamics, and shock physics.
- The scope is simply to include a listing of variables, equations, and methods used. It is not intended to be a substitute for textbooks on continuum mechanics and the finite element method.

Chapter 2

Formulation

2.1 Elastodynamics

The bulk modulus B , a measure of local compressibility of a substance, is defined as the ratio of infinitesimal change in pressure dP caused by an infinitesimal change in volume dV , at a given volume V

$$B \triangleq -V \frac{dP}{dV}. \quad (2.1)$$

We require $B > 0$, and since $V > 0$, then $dP/dV < 0$. This results in behaviors observed physically, namely that, all else being equal,

- increase in pressure causes decrease in volume, and vice versa, $\uparrow P \iff \downarrow V$, and
- increase in volume causes decrease in pressure, and vice versa, $\uparrow V \iff \downarrow P$.

Using the definition of density $\rho \triangleq m/V$,

$$\frac{d\rho}{dV} = \frac{-m}{V^2}, \quad (2.2)$$

and

$$\frac{dP}{dV} = \frac{dP}{d\rho} \frac{d\rho}{dV}, \quad (2.3)$$

the bulk modulus B may also be written as

$$B = \rho \frac{dP}{d\rho}. \quad (2.4)$$

Since $B > 0$ and $\rho > 0$, then $dP/d\rho > 0$. This result physically means that, all else being equal,

- increases in pressure cause increases in density (and vice versa), and
- increases in density cause increases in pressure (and vice versa).

2.2 Thermodynamics

2.2.1 Properties

Thermodynamics properties can be categorized as either **intensive** or **extensive**.

- Intensive properties are independent of, and therefore not proportional to, the size or sample of the material. Intensive properties are also referred to as **intrinsic** properties or, equivalently, as **bulk** properties.
- In contrast, extensive properties are dependent on, and therefore proportional to, the size or sample of the material. Extrinsic properties are also referred to as **extrinsic** properties.

Figure 2.1 demonstrates the independence of intrinsic properties and the dependence of extrinsic properties on the control volume, designated with the dashed lines. The original control volume, on the left side of the figure, is divided into two equal-sized control volumes, as shown on the right side of the figure. The intrinsic properties do not depend on the size of the control volume. The extrinsic properties, however, are now scaled by a factor of $\frac{1}{2}$.

Intrinsic (intensive, bulk) pressure, density, temperature	P, ρ, T	P, ρ, T	P, ρ, T
Extrinsic (extensive) mass, volume, energy	m, V, E	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$	$\frac{m}{2}, \frac{V}{2}, \frac{E}{2}$
Specific (extrinsic/mass) specific volume, specific energy	$\mathcal{V}, \mathcal{E}$	$\mathcal{V}, \mathcal{E}$	$\mathcal{V}, \mathcal{E}$

Figure 2.1: Thermodynamic properties categorized as intrinsic, extrinsic, and specific.

It is often convenient to speak of extrinsic properties in away that makes them *independent* of size or sample of the material. To achieve this, the extrinsic properties are divided by the mass of the sample. This ratio is referred to as the specific extrinsic property. Two common examples of specific properties are the **specific volume** and **specific energy**, defined as

$$\mathcal{V} \triangleq \frac{V}{m} = \frac{1}{\rho}, \quad (2.5)$$

and

$$\mathcal{E} \triangleq \frac{E}{m}. \quad (2.6)$$

2.2.2 First Law of Thermodynamics

The first law for a closed system states the change in internal energy $\Delta\mathcal{I}$ is equal to the net energy Q added to the system as heat less the net work W done **by** the system.

$$\Delta\mathcal{I} = Q - W. \quad (2.7)$$

This was the sign convention used by Joule and Clausius. A different sign convention, used by Planck, stated that the change in internal energy $\Delta\mathcal{I}$ is equal to the net energy Q added to the system as heat plus the net work done **on** the system.

$$\Delta\mathcal{I} = Q + W. \quad (2.8)$$

Thus, all net transfers **to** the system are **positive**. In contrast, all net transfers **from** the system are **negative**.

Then, the conservation of energy principle for open systems can be written as

$$\Delta E = \Delta(\mathcal{K} + \mathcal{I} + \mathcal{P}) = Q + W, \quad (2.9)$$

where the total energy E is the sum of the kinetic energy $\mathcal{K}$, the internal energy $\mathcal{I}$, and the potential energy $\mathcal{P}$. The change in total energy is equal to the net energy Q added to the system as heat plus the net work W done **on** the system.

Next, we introduce

- $d(\cdot)$ as the **exact differential**, which does not depend on the path taken through space of the thermodynamic parameters and depends only on the initial states, and
- $\delta(\cdot)$ as the **inexact differential**, which depends on the path taken through space of thermodynamic parameters.

The balance law then states

$$dE = \delta Q + \delta W. \quad (2.10)$$

2.2.3 Second Law of Thermodynamics

The Second Law states that the infinitesimal increment of entropy dS of a system from an infinitesimal transfer of heat δQ , divided by the common temperature of the system and the surrounding that supplies the heat, satisfies

$$dS \geq \frac{\delta Q}{T}. \quad (2.11)$$

For a reversible process,

$$dS = \frac{\delta Q}{T}. \quad (2.12)$$

For an irreversible process,

$$dS > \frac{\delta Q}{T}. \quad (2.13)$$

2.2.4 First and Second Laws Combined

For a reversible process, the first and second laws can be combined as

$$\boxed{dE = TdS + \delta W, \quad (\text{1st and 2nd Law, Reversible Form}).} \quad (2.14)$$

2.2.5 Thermodynamic Potentials

Variables

P	pressure	(N/m ²)
V	volume	(m ³)
T	temperature	(K)
S	entropy	(J/K)

Potentials

E	(?internal only?) energy	(J)
$F \triangleq E - TS$	Helmholtz free energy	(J)
$G \triangleq H - TS$	Gibbs free energy	(J)
$H \triangleq E + PV$	enthalpy	(J)

The **Helmholtz** free energy F is the Legendre transform of the (?internal?) energy E , where the entropy S is replaced by the temperature T as the independent variable.

Differential Forms of Potentials

dE	$=$	$T dS$	$-$	$P dV$	to come
dF	$=$	$-P dV$	$-$	$S dT$	xx
dG	$=$	$-S dT$	$+$	$V dP$	xx
dH	$=$	$V dP$	$+$	$T dS$	xx

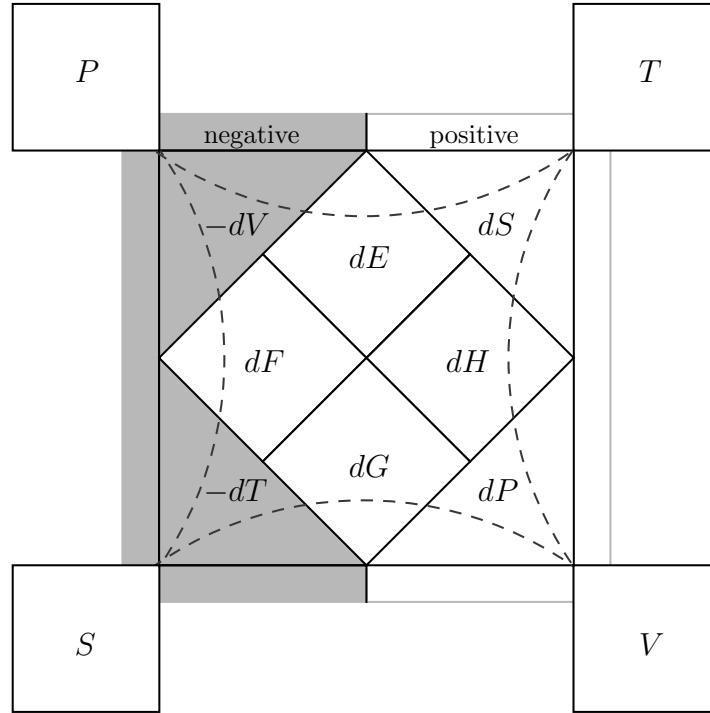


Figure 2.2: Mnemonic for the quick recall of the thermodynamic potentials in differential form.

2.2.6 Maxwell Relations

The Maxwell relations are a statement of equality between *first* derivative of a function because of a quality possessed by the *second* derivative of that function. Specifically, the Maxwell relations equate *first* derivatives of the thermodynamic potential functions E , F , G , and H , because the *second* derivatives of those same functions does not depend on the order of differentiation with respect to the thermodynamic variables P , V , T , and S .

The Maxwell relations make use of the **Schwarz theorem**, which states that if function $f : \mathbb{R}^2 \mapsto \mathbb{R}$ has continuous second partial derivatives at a given point

$\mathbf{p} \in \mathbb{R}^2 = \{p_1, p_2\}^T$,¹ then

$$\frac{\partial^2 f(\mathbf{p})}{\partial x_i \partial x_k} = \frac{\partial^2 f(\mathbf{p})}{\partial x_k \partial x_i} \quad (\text{Commutative}) \quad (2.15)$$

The Schwarz theorem establishes that the result of differentiation of a function of two variables is independent of the order of that differentiation.

With this context in mind, consider now the *first* derivatives of the thermodynamic potentials with respect to their natural variables. Specifically, we can write the four thermodynamic potentials as functions of their natural variables:

$$E = E(S, V), \quad (2.16)$$

$$F = F(V, T), \quad (2.17)$$

$$G = G(T, P), \quad (2.18)$$

$$H = H(P, S). \quad (2.19)$$

Next, we take *first* derivatives of these potentials:

$$\left(\frac{\partial E}{\partial S}\right)_V = T, \quad \left(\frac{\partial E}{\partial V}\right)_S = -P, \quad (2.20)$$

$$\left(\frac{\partial F}{\partial V}\right)_T = -P, \quad \left(\frac{\partial F}{\partial T}\right)_V = -S, \quad (2.21)$$

$$\left(\frac{\partial G}{\partial T}\right)_P = -S, \quad \left(\frac{\partial G}{\partial P}\right)_T = V, \quad (2.22)$$

$$\left(\frac{\partial H}{\partial P}\right)_S = V, \quad \left(\frac{\partial H}{\partial S}\right)_P = T. \quad (2.23)$$

Finally, we take *second* derivatives of the potential functions with respect to the natural variables, which allows for the *first* derivatives of the natural variables with respect to the natural variables themselves to be equated, as follows:

$$\frac{\partial^2 E}{\partial V \partial S} \implies \left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{-\partial P}{\partial S}\right)_V \quad (2.24)$$

¹This relationship can be extended to general dimensionality of $f : \mathbb{R}^n \mapsto \mathbb{R}$, but this higher dimensionality is not used in the development that follows.

$$\frac{\partial^2 F}{\partial T \partial V} \Rightarrow \left(\frac{-\partial P}{\partial T} \right)_V = \left(\frac{-\partial S}{\partial V} \right)_T \quad (2.25)$$

$$\frac{\partial^2 G}{\partial P \partial T} \Rightarrow \left(\frac{-\partial S}{\partial P} \right)_T = \left(\frac{\partial V}{\partial T} \right)_P \quad (2.26)$$

$$\frac{\partial^2 H}{\partial S \partial P} \Rightarrow \left(\frac{\partial V}{\partial S} \right)_P = \left(\frac{\partial T}{\partial P} \right)_S \quad (2.27)$$

The foregoing equations are useful for eliminating entropy S , which is difficult to measure in a laboratory setting, in favor of the other natural variables pressure P , volume V , and temperature T , all of which are relatively easier to measure.

2.3 Shock

2.3.1 Introduction

2.3.2 An Ideal Fluid

2.3.3 Definition of a Shock

2.3.4 Conservation Laws for a Single Shock

Laboratory Fixed Coordinate Frame

The derivation that follows assumes a **laboratory-fixed reference frame**,² as shown in Fig. 2.3. Consider a rigid tube containing a rigid piston and an ideal fluid at ambient density ρ_0 and ambient pressure P_0 at time $t = t_0$.

Assume at time $t = (t_0 + \Delta t)$, the piston moves left-to-right along the x -axis with speed u_P , measured in the fixed laboratory reference frame. Assume the piston speed u_P causes a shock to propagate through the material with speed U_S . We require the following:

$$0 < u_P < +\infty \quad (2.28)$$

$$0 < U_S < +\infty \quad (2.29)$$

$$u_P < U_S \quad (2.30)$$

From these considerations, we conclude that the material to the left of the shock wave in Fig. 2.3 moves to the right at speed u_p . The material to the right of the shock

²A shock-fixed reference frame derivation is developed in a subsequent section.

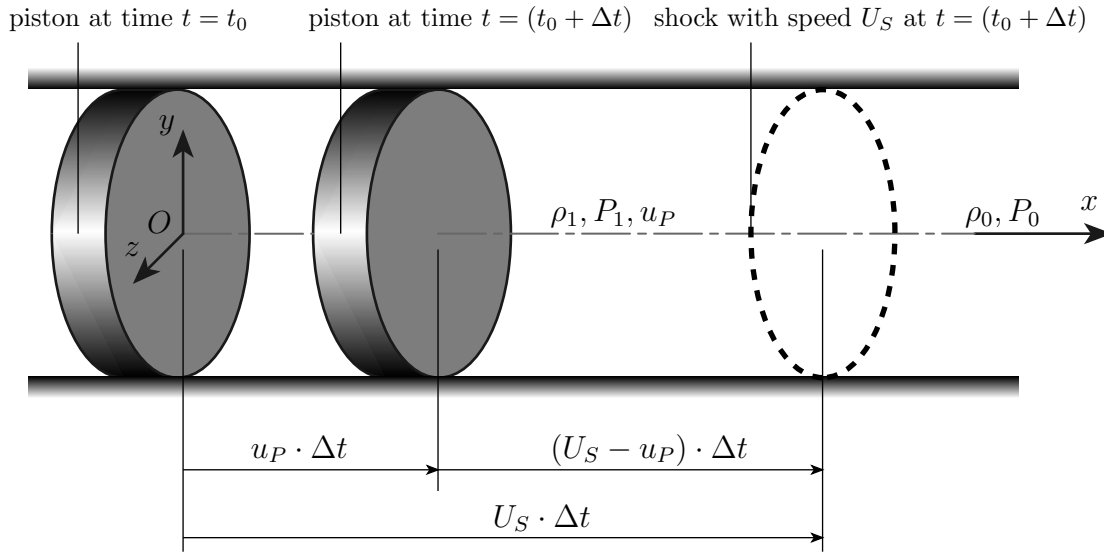


Figure 2.3: Piston moving left-to-right along the x -axis, with piston speed u_P , causing shock with shock speed U_S .

wave remains at rest.³

Since the piston compresses the material in the tube, we know

- $\rho_1 > \rho_0$,

where ρ_1 is the density of the material to the left of the shock.

Allow some time interval Δt to elapse, as illustrated in Fig. 2.3. Consider the distances along the x -axis, written as a multiplication of speeds and elapsed time. In the time interval Δt , the piston moves along the x -axis a distance of $u_P \cdot \Delta t$. In the same time interval, the shock wave moves along the x -axis a distance of $U_S \cdot \Delta t$. Thus, the control volume at $t = (t_0 + \Delta t)$ has an axial length of $(U_S - u_P) \cdot \Delta t$.

From these axial lengths and piston cross-sectional area A , we can write the following relationships:

- volume

$$V_0 = V|_{t=t_0} = A \cdot U_S \cdot \Delta t,$$

³Furthermore, we assume the material throughout the tube and for all time is inviscid, non-diffusive, and non-reacting. We assume an adiabatic boundary, thus no heat conduction occurs through the boundary of the tube walls or piston. We assume no heat transfer from radiation occurs across the shock. Finally, we assume body forces from gravity and electromagnetism are negligible.

$$V_1 = V|_{t=(t_0+\Delta t)} = A \cdot (U_S - u_P) \cdot \Delta t,$$

- density

$$\rho_0 = \rho|_{t=t_0},$$

$$\rho_1 = \rho|_{t=(t_0+\Delta t)},$$

- mass

$$\rho_0 V_0 = \rho_0 \cdot A \cdot U_S \cdot \Delta t,$$

$$\rho_1 V_1 = \rho_1 \cdot A \cdot (U_S - u_P) \cdot \Delta t.$$

Conservation of Mass

Conservation of mass within the control volume requires that mass at time $t = (t_0 + \Delta t)$ be unchanged from mass at time $t = t_0$. Thus

$$\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t = \rho_1 \cdot \mathcal{A} \cdot (U_S - u_P) \cdot \Delta t, \quad (2.31)$$

or

$$\boxed{\rho_0 U_S = \rho_1 (U_S - u_P), \quad \text{(Conservation of Mass).}} \quad (2.32)$$

Using specific volume (2.5), conservation of mass be rewritten as

$$\mathcal{V}_1 = \mathcal{V}_0 \left(1 - \frac{u_P}{U_S} \right), \quad (2.33)$$

or

$$\boxed{\mathcal{V}_1 - \mathcal{V}_0 = \frac{-v_0 u_P}{U_S}, \quad \text{(Conservation of Mass, Volume Form).}} \quad (2.34)$$

From (2.28)–(2.30), we know $\mathcal{V}_1 < \mathcal{V}_0$, which makes sense since we are compressing the material and generally have the understanding that increased pressures causes decreased volumes.

Conservation of Momentum

The net driving force from the piston on the material is $(P_1 - P_0) \cdot A$. Conservation of momentum requires $F = d(ma)/dt$. Written over the time interval Δt and for constant mass, conservation of momentum can be written

$$F \cdot \Delta t = m \cdot \Delta U. \quad (2.35)$$

Rewriting this expression in terms of pressures and densities, we have

$$(P_1 - P_0) \cdot \mathcal{A} \cdot \Delta t = (\rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t) (u_P - 0) \quad (2.36)$$

which gives

$$\boxed{P_1 - P_0 = \rho_0 U_S u_P, \quad \text{(Conservation of Momentum).}} \quad (2.37)$$

Using (2.5), (2.37) can be rewritten as

$$\boxed{P_1 - P_0 = \frac{U_S u_P}{\mathcal{V}_0}, \quad \text{(Conservation of Momentum, Volume Form).}} \quad (2.38)$$

Conservation of Energy

The work done on the control volume by the piston moving left-to-right is given by

$$W_P = \underbrace{P_1 \cdot A}_{\text{force}} \cdot \underbrace{u_P \cdot \Delta t}_{\text{disp.}} \quad (2.39)$$

Note that the right boundary of the control volume, located at the shock location, does not move, so the work across this boundary is identically zero.

The kinetic energy increase of the control volume is simply the one-half times the mass times the square of the change in velocity, thus

$$KE = \frac{1}{2} \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \underbrace{(u_P - 0)^2}_{\text{velocity}} \quad (2.40)$$

Finally, the increase in internal energy is just the internal energy **per unit mass**, $\mathcal{E}$, multiplied by the mass,

$$IE = (\mathcal{E}_1 - \mathcal{E}_0) \underbrace{(\rho_0 \cdot A \cdot U_S \cdot \Delta t)}_{\text{mass}} \quad (2.41)$$

Equating these quantities gives

$$P_1 \cdot \mathcal{A} \cdot u_P \cdot \Delta t = \frac{1}{2} \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \cdot u_P^2 + (\mathcal{E}_1 - \mathcal{E}_0) \rho_0 \cdot \mathcal{A} \cdot U_S \cdot \Delta t \quad (2.42)$$

or simply

$$\boxed{P_1 u_P = \rho_0 U_S \left(\frac{u_P^2}{2} + \mathcal{E}_1 - \mathcal{E}_0 \right), \quad \text{(Conservation of Energy).}} \quad (2.43)$$

The foregoing equation can be written without kinematic variables U_S and u_P . Next, note that conservation of mass can be rewritten as

$$\frac{u_P}{U_S} = 1 - \frac{\rho_0}{\rho_1}. \quad (2.44)$$

Then, with (2.43),

$$P_1 u_P \stackrel{(2.37)}{=} (P_1 - P_0) \frac{u_P}{2} + \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (2.45)$$

$$= -P_0 u_P + 2 \rho_0 U_S (\mathcal{E}_1 - \mathcal{E}_0), \quad (2.46)$$

$$(\mathcal{E}_1 - \mathcal{E}_0) = \frac{u_P (P_1 + P_0)}{2 \rho_0 U_S}, \quad (2.47)$$

$$\stackrel{(2.44)}{=} \left(\frac{P_1 + P_0}{2 \rho_0} \right) \left(1 - \frac{\rho_0}{\rho_1} \right), \quad (2.48)$$

$$\stackrel{(2.5)}{=} \left(\frac{P_1 + P_0}{2/\mathcal{V}_0} \right) \left(1 - \frac{\mathcal{V}_1}{\mathcal{V}_0} \right), \quad (2.49)$$

which simplifies to

$$\boxed{\mathcal{E}_1 - \mathcal{E}_0 = \frac{-1}{2} (P_1 + P_0) (\mathcal{V}_1 - \mathcal{V}_0), \quad \textbf{(Rankine-Hugoniot curve).}} \quad (2.50)$$

The Rankine-Hugoniot equation is the Conservation of Energy jump condition with state variables alone; there are no kinematic variables in the equation. To plot the Rankine-Hugoniot relationship in the $P\mathcal{V}$ plane with $\mathcal{E}_1 > \mathcal{E}_0$, the initial state $(\mathcal{V}_0, P_0)$ must be known.

Speed-Volume-Pressure Relationships

The alternative forms for mass conservation (2.34) and momentum conservation (2.38), when multiplied together or divided by one another, readily allow for the shock speed U_S and particle speed u_P to be written in terms of volume and pressure. These relationship are found to be

$$U_S = \mathcal{V}_0 \sqrt{\frac{P_1 - P_0}{-(\mathcal{V}_1 - \mathcal{V}_0)}} = \mathcal{V}_0 \sqrt{\frac{\Delta P}{-\Delta \mathcal{V}}}, \quad (2.51)$$

and

$$u_P = \sqrt{-(P_1 - P_0) (\mathcal{V}_1 - \mathcal{V}_0)} = \sqrt{-\Delta P \Delta \mathcal{V}}. \quad (2.52)$$

Alternative Formulations

Now we develop the shock-fixed reference frame equations for mass, momentum, and energy conservation. (To come.)

2.3.5 $P\mathcal{V}$ Diagram

The $P\mathcal{V}$ diagram shows pressure P as a function of specific volume $\mathcal{V}$. Thus, the state variables P and $\mathcal{V}$ are considered in isolation from the other state and kinematic variables. The initial laboratory state is denoted $(\mathcal{V}_0, P_0)$. A large compressive pressure P_1 is applied to the material, which causes the volume to shrink to $\mathcal{V}_1$. If this applied pressure P_1 causes a shock, we record the pressure and volume state, and then repeat the experiment at a different pressure, e.g., P_2 . Several compression experiments are run, and the $P\mathcal{V}$ states that created a shock in the material are recorded in the $P\mathcal{V}$ diagram.

For concreteness, we show four experiments with applied pressures $P_1 \dots P_4$ and resultant volumes $\mathcal{V}_1 \dots \mathcal{V}_4$, shown in Fig. 2.4. Note that all experiments start from the same initial pressure P_0 and initial specific volume $\mathcal{V}_0$.

These four thermodynamic states lie on a locus of all possible thermodynamic states that can exist in the material behind a shock. This locus is referred to as the **shock Hugoniot**. The curve connecting the locus is the **Rankine-Hugoniot curve**. The line connecting two end states is the **Rayleigh line**.

It is important to emphasize that the Rankine-Hugoniot curve **is not** the path taken by the material between two thermodynamic states.

Note that (2.51) describes the mathematical increase in shock speed with increasing slope of the Hugoniot, indicated in Fig. 2.4. Specifically, the slope of the Hugoniot goes as the negative squared value of the shock speed U_S per initial specific volume $\mathcal{V}_0$, or

$$\frac{\Delta P}{\Delta \mathcal{V}} = - \left(\frac{U_S}{\mathcal{V}_0} \right)^2. \quad (2.53)$$

2.3.6 Pu_P Diagram

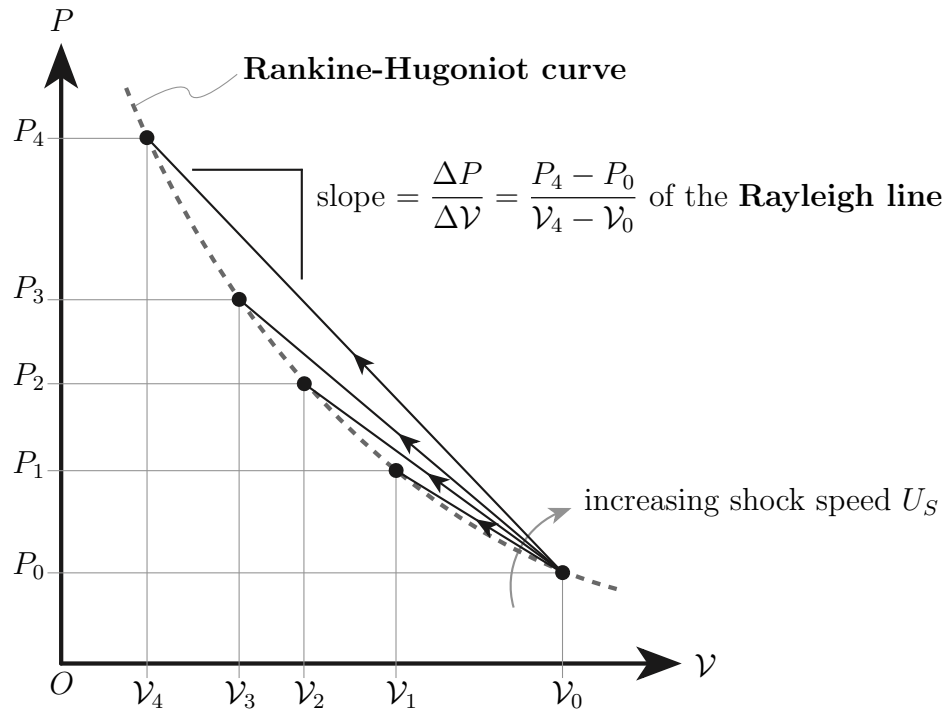


Figure 2.4: Pressure as a function of volume for the shock Hugoniot locus of thermodynamic equilibrium states.

Chapter 3

Implementation